

Reg No.: \_\_\_\_\_

Name: \_\_\_\_\_

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**

B.Tech Degree S3 (S,FE) / S1 (PT)(S) June 2024 (2019 Scheme)/S3 (WP)(R) December 2023 Examination

**Course Code: MAT201****Course Name: Partial Differential Equations and Complex Analysis**

Max. Marks: 100

Duration: 3 Hours

**PART A***Answer all questions. Each question carries 3 marks*

Marks

- 1 Form the partial differential equation by eliminating arbitrary function from the relation  $z = x - y + f(x^2 + y^2)$ . (3)
- 2 Solve by direct integration:  $\frac{\partial^2 z}{\partial x \partial y} = \sin(3x + 4y)$ . (3)
- 3 Write all possible solutions of one-dimensional wave equation. (3)
- 4 Write any three assumptions used in the derivation of one-dimensional heat equation. (3)
- 5 Check whether  $f(z) = z - \bar{z}$  is analytic or not. (3)
- 6 Test the continuity of  $f(z)$  at the point  $z = 0$ , where  $f(z) = \begin{cases} \frac{\operatorname{Im} z}{|z|}, & z \neq 0 \\ 0, & z = 0 \end{cases}$  (3)
- 7 Find the value of the integral  $\int_C z \, dz$  where  $C$  is the part of the unit circle from  $-i$  to  $i$  in the right half plane. (3)
- 8 Evaluate  $\oint_C \frac{\cos z}{z^2 + 9} dz$ , where  $C$  is the circle  $|z| = 1$ . (3)
- 9 Find the Laurent series expansion of  $\frac{1}{1-z^2}$  about  $z=0$  (3)
- 10 Find the singularities and nature of singularities of the function  $f(z) = z^2 \cos \frac{1}{z}$  (3)

**PART B***Answer any one full question from each module. Each question carries 14 marks***Module 1**

- 11(a) Solve the PDE  $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$ . (7)
- (b) Solve by Method of Separation of variable  $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial t} = 0, u(x, 0) = 4e^{-3x}$ . (7)
- 12(a) Solve  $2xz - px^2 - 2qxy + pq = 0$ . (7)
- (b) Solve  $(y^2 + z^2)p - xyq = xz$  (7)

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### Module 2

- 13(a) A tightly stretched string with fixed end points. One end is at  $x = 0$  and the other at  $x = l$  is initially at rest in its equilibrium position. If it is vibrating by giving each of its points a velocity  $\lambda x(l - x)$ . Find the displacement of the string at any distance  $x$  from one end at any time  $t$ . (7)
- (b) Derive One Dimensional Wave equation. (7)
- 14(a) Derive all possible solutions of one-dimensional wave equation. (7)
- (b) Find the temperature  $u(x, t)$  in a homogeneous rod of length  $2m$ , whose ends are kept at  $0^\circ\text{C}$  and whose initial temperature is given by  $f(x) = 100(2x - x^2)$ ,  $0 \leq x \leq 2$  (7)

### Module 3

- 15(a) Show that  $f(z) = e^z$  is analytic for all  $z$  in the complex plane and find its derivative. (7)
- (b) Find image of the circle  $\left|z - \frac{1}{2}\right| \leq \frac{1}{2}$  under the transformation  $w = \frac{1}{z}$ . (7)
- 16(a) Show that the function  $u = \cos x \cosh y$  is harmonic and find its harmonic conjugate. (7)
- (b) Show that an analytic function  $f(z) = u + iv$  is constant if  $|f(z)|$  is constant (7)

### Module 4

- 17(a) Evaluate  $\int_C \frac{2z+1}{z^2-2z} dz$  where  $|z| = 1$ . (7)
- (b) Find the Taylor series expansion of  $f(z) = \sin 2z$  about  $z = \frac{\pi}{2}$ . (7)
- 18(a) Evaluate  $\int_C |z| dz$  where  $C$  is the part of the circle  $|z| = 3$  from  $3$  to  $-3$  in the (7)
- (b) upper half plane. (7)
- Evaluate  $\int_C \frac{e^{3z}}{(z+1)^4} dz$  where  $C$  is the circle  $|z - 1| = 3$ .

### Module 5

- 19(a) Find Laurent series expansion  $f(z) = \frac{-2z+3}{(z^2-3z+2)}$  valid in (7)
- i.  $1 \leq |z| \leq 2$
- ii.  $|z| \geq 2$
- (b) Evaluate  $\oint_C \frac{e^z}{z^2(z+1)} dz$ , over the circle  $|z| = 2$  (7)
- 20(a) Evaluate  $\int_C \tan z dz$  where  $C$  is circle  $|z| = 2$ . (7)
- (b) Evaluate  $\int_0^{2\pi} \frac{d\theta}{\sqrt{2}-\cos \theta}$  (7)